In group 14, the first element carbon is strictly a nonmetal while other elements are metalloids (silicon & germanium) or metals (tin & lead). Unlike other elements of the group carbon can form multiple bonds such as C=C, C=O etc... Carbon has a greater tendency to form a chain of bonds with itself or with other atoms which is known as catenation. There is considerable decrease in catenation property down the group (C>>Si>Ge≈Sn>Pb).

In group 15 also the first element nitrogen differs from the rest of the elements of the group. Like carbon, the nitrogen can from multiple bonds (N=N, C=N, N=O etc...). Nitrogen is a diatomic gas unlike the other members of the group. Similarly in group 16, the first element, oxygen also exists as a diatomic gas in that group. Due to its high electronegativity it forms hydrogen bonds.

The first element of group 17, fluorine the most electronegative element, also behaves quiet differently compared to the rest of the members of group. Like oxygen it also forms hydrogen bonds. It shows only -1 oxidation state while the other halogens have +1, +3, +5 and +7 oxidation states in addition to -1 state. The fluorine also is the strongest oxidising agent and the most reactive element among the halogens.

2.1.6 Inert pair effect:

We have already learnt that the alkali and alkaline earth metals have an oxidation state of +1 and +2 respectively, corresponding to the total number of electrons present in them. Similarly, the elements of p-block also show the oxidation states corresponding to the maximum number of valence electrons (group oxidation state). In addition they also show variable oxidation state. In case of the heavier post-transition elements belonging to the groups (13 to 16), the most stable oxidation state is two less than the group oxidation state and there is a reluctance to exhibit the group oxidation state. Let us consider group 13 elements. As we move from boron to heavier elements, there is an increasing tendency to have +1 oxidation state, rather than the group oxidation state, +3. For example Al^{+3} is more stable than Al^{+1} while Tl^{+1} is more stable than Tl^{+3}. Aluminium(III)chloride is stable whereas thallium(III)chloride is highly unstable and disproportionates to thallium(I) chloride and chlorine gas. This shows that in thallium the stable lower oxidation state corresponds to the loss of np electrons only and not ns electrons. Thus in heavier post-transition metals, the outer s electrons (ns) have a tendency to remain inert and show reluctance to take part in the bonding, which is known as inert pair effect. This effect is also observed in groups 14, 15 and 16.

2.2.7 Allotropism in p-block elements:

Some elements exist in more than one crystalline or molecular forms in the same physical state. For example, carbon exists as diamond and graphite. This phenomenon is called allotropism (in greek *'allos'* means another and '*trope'* means change) and the different forms of an element are called allotropes. Many p-block elements show allotropism and some of the common allotropes are listed in the table.

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